



2.4 Dilated and Restrictive Cardiomyopathies

1) Beat up hearts:

cardiomyopathy (primary pathology of the CARDIAC MUSCLE, rather than pathology secondary to HTN, CAD, valvular dz, etc.)

2) Failing heart balloon:

Most **cardiomyopathies** manifest clinically with symptoms & signs of heart failure (e.g., dyspnea on exertion, peripheral edema, inspiratory crackles, elevated JVP, S3 gallop)

Dilated Cardiomyopathy

3) Dilated heart sack with floppy accordion:

Dilated cardiomyopathy (DCM) = dilation AND impaired contraction (i.e. systolic dysfunction) of the LV or both ventricles

4) Struggling for air:

cardiomyopathy (DCM, RCM) manifests clinically with symptoms of heart failure (e.g. dyspnea on exertion, fatigue) (**Chief Complaint**)

5) Dilated wet socks:

cardiomyopathy (DCM, RCM) manifests clinically with symptoms of heart failure (e.g. peripheral edema) (**Chief Complaint**)

6) Flaming heart:

Myocarditis (e.g. **viral**) is the most common known cause of **DCM**

7) Blowing nose with flaming heart:

Myocarditis may be associated with recent **viral prodromal symptoms** (e.g., fever, malaise, myalgias) preceding symptoms of decompensated HF (**HPI**)

8) Angina anvil with flaming heart:

Myocarditis with pericardial involvement (**myopericarditis**) may present with chest pain that resembles angina (**HPI**)

9) Stressed sack racer:

Stress-induced cardiomyopathy (Takotsubo cardiomyopathy) is an increasingly reported cause of **DCM** → ask about recent emotional stress (**HPI**)

10) Angina anvil with stressed sack racer:

Stress-induced cardiomyopathy may present with chest pain that resembles angina (**HPI**)

38) Elevated ST street sign:

Stress-induced cardiomyopathy and **myocarditis** can presents with ST elevations on ECG (both conditions will reveal a NORMAL coronary angiogram) (**ECG**)

43) Ballooning octopus pattern:

stress induced cardiomyopathy features “apical ballooning” that looks like an “octopus trap” on echo or ventriculogram

11) Iron weight with dilated heart sack:

hereditary hemochromatosis can cause **DCM** (may also cause skin hyperpigmentation, diabetes, liver disease/cirrhosis) (**PMHx**)

12) Soccer ball with dilated heart sack:

DCM is a late cardiac manifestation of **sarcoidosis** (may also cause arrhythmias, conduction abnormalities (classically 3rd degree AV block), bilateral hilar lymphadenopathy) (**PMHx**)

13) Sack racing mom holding baby:

peripartum cardiomyopathy (**DCM** can occur in last trimester, first few months postpartum (most common), up to 5 months postpartum) → take an OB history) (**PMHx**)

14) Radiation light on table:

Radiation therapy to the chest may lead to **DCM** or **RCM** (acutely → **pericarditis**) (years later → pericardial disease, CAD, valvular disease, cardiomyopathy) (**HPI**)

48) Radiation light on field:

Radiation therapy to the chest may lead to **DCM** or **RCM** (acutely → **pericarditis**) (years later → pericardial disease, CAD, valvular disease, cardiomyopathy) (**HPI**)

15) Box-o'-rubies:

Use of anthracyclines (e.g. doxorubicin) is associated with **DCM** (**Meds**)

16) Cardiac tapestry:

Trastuzumab (used in Tx of HER2/neu+ breast cancer) is associated with **DCM** (**Meds**)

17) Medication prize bag:

Many **drugs** can cause **DCM** including anthracyclines (e.g. doxorubicin), trastuzumab (and other chemotherapeutic agents), ART therapy for HIV, phenothiazines (e.g. thorazine, prochlorperazine) (**Meds**)

18) Drinking in heart sack:

Alcoholic cardiomyopathy = diagnosis of exclusion in patients with **DCM** & history of alcohol abuse (**Social Hx**)

19) Broken drinking horn:

Complete cessation of alcohol intake can reverse **Alcoholic cardiomyopathy** (**Plan**) (1)

20) Berries! Berries!:

Chronic **thiamine (vitamin B1) deficiency** can cause **DCM** (aka wet beriberi) (**Social Hx**)

21) Cocoa:

Chronic cocaine abuse (and methamphetamines) can cause **DCM** (due to direct toxic effect and cardiac stress from prolonged hyperadrenergic state) (**Social Hx**)

22) Traveler under "C-H--G-A-S" sign:

chronic *Trypanosoma cruzi* infection (Chagas disease) can cause **DCM** (and conduction abnormalities, megaesophagus, megacolon) → ask about travel history (most common cause of DCM in Central and South America) (**Social Hx**)

23) "Idiot" sack racer:

idiopathic DCM (~half of all cases)

24) Family at sack race:

DCM may have a familial component (~half of all idiopathic cases; also associated with HH, **muscular dystrophy**, **myotonic dystrophy**, and **Friedreich's ataxia**) (**Family Hx**)

25) Elevated heart watch at sack race:

severe persistent tachycardia can cause **DCM** (e.g persistent asymptomatic AFib, AFlutter, or SVT with heart rates > 130) (**Vitals/ECG**)

26) Wet chest:

cardiomyopathy (**DCM**, RCM) manifests clinically with signs of heart failure (e.g. inspiratory crackles) (**Physical Exam**)

27) Edematous socks:

cardiomyopathy (**DCM**, RCM) manifests clinically with signs of heart failure (e.g. peripheral edema) (**Physical Exam**)

28) Dilated braids:

cardiomyopathy (**DCM**, RCM) manifests clinically with signs of heart failure (e.g. elevated JVP)(**Physical Exam**)

29) S3 "Slush3":

cardiomyopathy (**DCM**, RCM) manifests clinically with signs of heart failure (e.g. S3 gallop) (**Physical Exam**)

30) Regurging with systolic spray:

DCM can present with a blowing holosystolic murmur at the cardiac apex (due to 2° mitral regurg) (**Physical Exam**)

31) Displaced heart pendant:

A laterally displaced PMI suggests LV dilation which can distinguish **DCM** from other cardiomyopathies (**Physical Exam**)

32) Working up the failing heart balloon:

work up heart failure (associated with **cardiomyopathy**) with a CBC, BMP, BNP, tropes, EKG, CXR, and echo (**Labs/ECG/CXR/Echo**)

33) BNP blimp:

cardiomyopathy (**DCM**, RCM) can cause an **elevated BNP** (non-specific for HF of any cause)(**Labs**)

34) T-bones with flame sack and stressed sack:

Elevated troponins in the context of **DCM** suggests **myocarditis** or **stress-induced cardiomyopathy (Labs)**

35) Checking iron weight:

Abnormal iron studies (e.g. elevated transferrin saturation and serum transferrin levels) in the context of **DCM** suggests **hereditary hemochromatosis (Labs)**

36) ECG tartan at the sack race:

on ECG, **DCM** can present with AV block (1st, 2nd, 3rd degree), intraventricular conduction defects (e.g. LBB block, LAF block, QRS prolongation), ST elevations, and tachyarrhythmias (**ECG**)

37) Blocking heart shield at the sack race:

DCM can present with 1st degree AV block (may progress to 2nd or 3rd degree block esp. in cardiac infiltrative disease e.g. sarcoidosis)(**ECG**)

40) Ultrasounding bull horn at the sack races:

Echocardiography = diagnostic test of choice for **DCM** → look for dilation of atria and ventricles, decreased ventricular wall thickness, reduced EF, MR, and absence of other causes of heart failure (e.g., valvular disease, pulmonary HTN, intracardiac shunt) (non-diagnostic early) (**Echo**)

41) Dilated atrial balloons and ventricular body on thin wall:

On echo, **DCM** can present with dilation of atria and ventricles, with decreased ventricular wall thickness (**Echo**)

42) Falling cake fraction at the sack races:

On echo, **DCM** can present with decreased ejection fraction (**Echo**)

44) **Dilated Cardiomyopathy** DDx clipboard:

For further workup of Dilated Cardiomyopathy, see Chronic HF SOA

45) Patch on failing heart balloon:

DCM Treatment = heart failure medications (diuretics, beta-blockers, ACEi), and treat the underlying cause (**Plan**) (2)

Restrictive Cardiomyopathy

4) Struggling for air:

cardiomyopathy (DCM, **RCM**) manifests clinically with symptoms of heart failure (e.g. dyspnea on exertion, fatigue) (**Chief Complaint**)

5) Dilated wet socks:

cardiomyopathy (DCM, **RCM**) manifests clinically with symptoms of heart failure (e.g. peripheral edema) (**Chief Complaint**)

26) Wet chest:

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28) Dilated braids:

cardiomyopathy (DCM, **RCM**) manifests clinically with signs of heart failure (e.g. elevated JVP) (**Physical Exam**)

29) S3 "Slushi3":

cardiomyopathy (DCM, **RCM**) manifests clinically with signs of heart failure (e.g. S3 gallop) (**Physical Exam**)

46) Restricted heart net causing filling problems:

Restrictive cardiomyopathy (**RCM**) = impaired ventricular filling (i.e. diastolic dysfunction), decreased ventricular compliance, typically normal ventricular walls (no hypertrophy or dilation)

47) Crushed RIGHT side at the heart net:

RCM manifests clinically with SYMPTOMS of HF (RIGHT HF more common → peripheral edema, abdominal distention) (**Chief Complaint**)

48) Radiation light on field:

Radiation therapy to the chest may lead to **DCM** or **RCM** (acutely → **pericarditis**) (years later → pericardial disease, CAD, valvular disease, cardiomyopathy) (**HPI**)

49) Armored lady with aggregated chainmail:

amyloidosis can cause **RCM** (rarely DCM) (other systemic manifestations include proteinuria, neuropathy, hepatomegaly) (**PMHx**)

50) Soccer ball in heart net:

Sarcoidosis can cause **RCM** (→DCM late) (may also cause arrhythmias, conduction abnormalities (classically 3rd degree AV block), bilateral hilar lymphadenopathy) (**PMHx**)

51) Iron weight in heart net:

hereditary hemochromatosis can cause **RCM** (may also cause skin hyperpigmentation, diabetes, liver disease/cirrhosis) (**PMHx**)

52) Sclerotic dragon:

systemic sclerosis can cause **RCM** (DCM less common) (may also cause joint pain, dysphagia and skin thickening) (**PMHx**)

53) Distended blue braids with falling Y glass:

RCM can present with signs of **RIGHT HF** → elevated JVP with prominent Y descent (**Physical Exam**)

54) Distended blue braids with cookie smell:

RCM can present with signs of **RIGHT HF** → Kussmaul's sign (INCREASE in JVP with inspiration) (**Physical Exam**)

55) Smacking liver with distended blue braids:

RCM can present with signs of **RIGHT HF** → Hepatojugular reflux (sustained JVP >3 cm for 15 seconds after pressing on the abdomen) (**Physical Exam**)

56) Signs of crushed RIGHT heart balloon:

RCM manifests clinically with other signs of **RIGHT HF** → hepatomegaly, ascites and peripheral edema (**Physical Exam**)

58) ECG tartan at the heart net:

RCM can present with variable ECG findings (e.g. AV block with infiltrative diseases, or LOW voltage readings with amyloidosis) (**ECG**)

59) Blocking heart shield at heart net:

RCM can present with AV block with infiltrative diseases (e.g. sarcoidosis, amyloidosis) (**ECG**)

60) Ultrasounding bull horn at the heart net:

Echocardiography = diagnostic test of choice for **RCM** → dilation of atria, preserved EF (amyloidosis shows biventricular wall thickening with granular texture) (**Echo**)

61) Dilated atrial balloons and thin ventricular body:

On echo, **RCM** can present with bi-atrial enlargement with a normal left ventricular thickness (may be thickened in infiltrative diseases) (**Echo**)

62) Preserved cake fraction at the heart net:

On echo, **RCM** can present with PRESERVED ejection fraction (**Echo**)

63) Restrictive Cardiomyopathy DDx clipboard:

For further workup of **Restrictive Cardiomyopathy**, see Pericardial Effusion, Cardiac Tamponade, & Constrictive Pericarditis SOAP

To Heart Failure

39) Wet skull and X-bones flag:

workup for **heart failure** includes **CXR**→ may see cardiomegaly (due to dilated heart chambers) and pulmonary vascular congestion (**non-specific to cause of cardiomyopathy**) (**CXR**)

57) Working up the failing heart balloon:

work up **heart failure** (associated with cardiomyopathy) with a CBC, BMP, BNP, tropo, EKG, CXR, and echo(**Labs/ECG/CXR/Echo**)